Universidad Internacional del Ecuador (UIDE)

Computer Science School ***Network Communication Technology***

Title: *GTA1_Phase_1_Integrated Project*

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1. Scenario and scope

Sites

As a team, we have proposed the development of this project in a distributed corporate environment, consisting of two physically separate organizational units, which we will call "Site A" and "Site B." In our main scenario, it is required to securely and efficiently interconnect both locations to support the corporate chat application we are developing, which is based on a client-server architecture. We are going to design and verify the entire network infrastructure that interconnects these sites in a structured simulation environment with Cisco Packet Tracer.

Actors and users

When designing the network ecosystem, we have identified two main categories of actors within the organization:

- → **Internal users (clients):** These are the collaborators and operators of the organization, physically located in Site A and Site B. These users will need continuous connection to the network to authenticate, send and receive text messages simultaneously and see their session history in the system.
- → **System / net administrator:** It is the technical role that takes care of the technology infrastructure. We define their responsibilities on the network to include auditing connection logs, managing the lifecycle of corporate user accounts, configuring

firewall rules, checking the availability of VLANs, and administering chat application services.

Physical and logical components of the reference architecture

In our simulation, we will implement the following logical and physical components to ensure smooth communication between both the locations:

- **End user devices:** User equipment (desktop or laptop computers) located at both sites. Interaction with the system takes place via an interactive client program running on the console or terminal of these devices.
- **Centralized servers:** We will deploy a dedicated server infrastructure to host the heart of our client-server chat in Go (handling sockets and concurrency) and the server of our authentication service written in FastAPI.
- **Switching equipment (switches):** Layer 2 devices (deployed in pairs at each site, such as SW-A1/SW-A2 and SW-B1/SW-B2) provide logical segmentation via VLANs (VLAN 10 for users, VLAN 20 for servers, VLAN 30 for guests, and VLAN 99 for management). Configurations include IEEE 802.1Q tagged trunks, Layer 2 redundancy to prevent loops using RSTP, and link aggregation via EtherChannel (LACP/PAgP).
- **Routing equipment (routers / OpenWrt):** OpenWrt-based routers (installed on Raspberry Pi-class gateways at each site) that route traffic between VLANs using the Router-on-a-Stick technique. Additionally, they interconnect Site A and Site B

via a shared segment (Wi-Fi/WAN in Area 0), using the OSPFv2 dynamic routing protocol to automatically exchange routes.

Core services

We will construct our network for the internal and centralized services that we have imagined for this project, not for deployment on commercial cloud platforms or public messaging gateways. The main services that will be running on our infrastructure are:

- → **Concurrent messaging service:** We will develop a central service in Go capable of efficiently managing multiple incoming network connections and relaying text packets in real time between client sockets.
- → **Authentication and authorization service:** A security service (API) we created with the FastAPI framework. This service will handle network login requests, verify password integrity using secure hashes, and provide authorization statuses to restrict access.
- → **Gateway network services:** Implementation in OpenWrt of NAT (independent Internet access per site), VLAN-based DHCP, firewall (traffic filtering), and FRRouting (OSPF) routing.

Communication and operational needs

Given the problems identified in the company, we have organized the network according to the following operational needs:

- **Centralization and organization:** Replace informal calls and emails with a shared internal platform that allows team members to coordinate activities without delays or loss of information.
- **Segmentation and security:** We will make sure that the company's traffic is logically isolated by means of VLANs and access to messaging is protected with strict validation of credentials. We will also establish firewall rules to block unauthorized departmental communications on the network.
- **Reliability and availability:** The continuous operation of the service will be ensured through the establishment of redundant links and dynamic routing (OSPF). This will ensure that messages will arrive at their destination without loss between the two locations. Furthermore, we will limit data transfer to text strings and lightweight logs, avoiding the transmission of multimedia or large files to optimize network performance and latency.

2. Requirements identification

To allow our application to function, be secure and operate correctly in a client server architecture we will implement the following protocols and technologies within the topology:

- **OpenWrt (Wireless access layer):** We will add SOHO devices in Access Point mode to extend the access layer and simulate a realistic corporate environment. These devices will be deployed at the edge of the network to provide connectivity

to the mobile nodes using WPA2-Enterprise encryption and bridging the wireless traffic directly to the associated wired VLANs.

- **VLANs (Layer 2 logical segmentation):** Virtual Local Area Networks will be used to physically segment the infrastructure and contain the broadcast domains. This segmentation will allow us to strictly isolate the operational traffic between departments and have a dedicated network segment that will host only the concurrent traffic of the Go and FastAPI servers.
- **VLSM:** For the IPv4 addressing we will follow a hierarchical design based on CIDR (Classless Inter Domain Routing). We will compute subnets with variable length prefixes mathematically adapted to the exact number of hosts per segment. We will use wide masks (/24 or /25) for client VLANs and /30 masks for point-to-point links between routers . This will optimize the IP address space without waste.
- **IEEE 802.1Q trunks:** The switching ports will be set to trunk mode for interconnection of the distribution and access equipment. This will inject a 4-byte tag in the original Ethernet frames. This will securely multiplex the traffic of multiple VLANs over unique physical links, without the packets losing their original network identifier.
- **Router-on-a-Stick:** We will accomplish Layer 3 inter-VLAN routing by breaking down the physical Gigabit Ethernet interface of our perimeter router into several logical subinterfaces (GigabitEthernet0/0.10). dot1Q encapsulation will be performed on each subinterface and will act as the Default Gateway for its subnet,

so that the client traffic can reach the central servers based on the routing policies in place.

- **EtherChannel:** To avoid bottlenecks toward the server in Go and high availability, we will configure logical port channels using the LACP (802.3ad) negotiation protocol. This will combine the bandwidth of several physical links into a single logical channel and balance the load of concurrent traffic with algorithms based on MAC or IP addresses.
- **RSTP:** In an effort to prevent Layer 2 switching loops caused by the redundant links made possible by EtherChannel, RSTP will be enabled. In case of any type of topology failure, RSTP will nearly instantaneously recalculate the roles of the ports, moving into a forwarding state in milliseconds, to avoid interruption of active sessions or packet transfer in the chat.
- **OSPF:** We will use OSPF as our interior gateway protocol (IGP). The routers at site A and site B will make adjacencies using hello packets and then run the Dijkstra algorithm to compute metrics based on the cost (bandwidth) of the links. This will provide dynamic routing (link state) with routing tables that are constantly updated, and fast convergence when WAN outages occur.
- **DHCP:** We will centralize IP management by creating DHCP pools on our layer 3 devices. The network settings (IP, subnet mask, gateway and DNS) will be dynamically assigned to end clients through the DORA process (Discover, Offer, Request, Acknowledge) excluding previously static addresses reserved for servers and router interfaces.

- → **NAT:** We'll configure NAT Overload on the outbound ports, only if our simulation goes through a generic ISP cloud, since our application will be working within an internal and isolated architecture. This will hide our private IPs behind a public IP and give transparent external connectivity."
- → **Firewall:** The system administrator will implement perimeter and internal security using extended ACLs on the critical interfaces of the routers. These rules will act as a stateless firewall, filtering on source/destination IP, and a specific TCP port. This is what we will be doing an "implicit deny" policy. Only packets to the specified Go and FastAPI ports will be allowed and all other traffic will be denied.

3. Preliminary topology

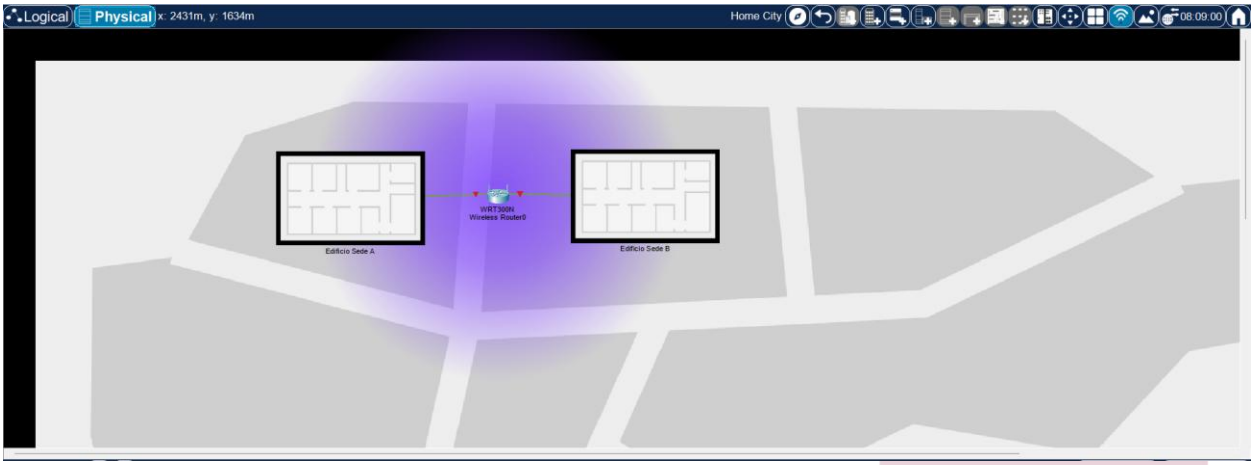
Physical Organization

The physical network layout has been structured hierarchically within Cisco Packet Tracer, ranging from the general inter-site view to the specific hardware implementation in the communications racks at each location. The environment connects Headquarters Building A and Headquarters Building B via a centralized, shared wireless transmission medium (Wireless Router0), simulating an external wide-area network (WAN) link that enables independent communication and dynamic routing between the two sites.

Within each building, end-user devices (desktop and laptop computers) connect via structured cabling to the equipment rooms (wiring closets), which house the racks containing power distribution units, edge routers, switches configured for redundancy, and dedicated servers (the Go and FastAPI cluster at Site A and the support node at Site B).

Figure 1

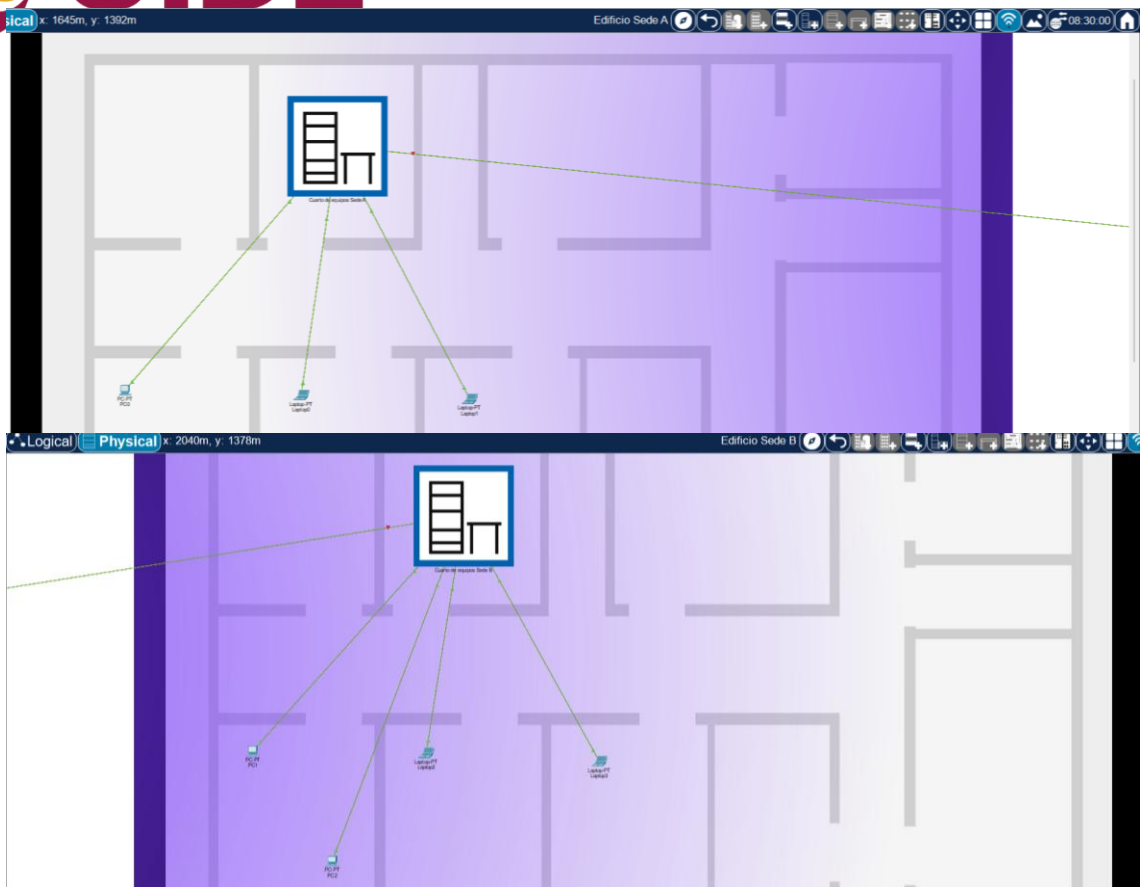
General physical view of Sites A and B interconnected by the central wireless router.



Note. Physical diagram in Cisco Packet Tracer showing the spatial location of Site A and Site B buildings within the city, linked via Wireless Router0, which operates as the central access point for the shared wireless network.

Figure 2

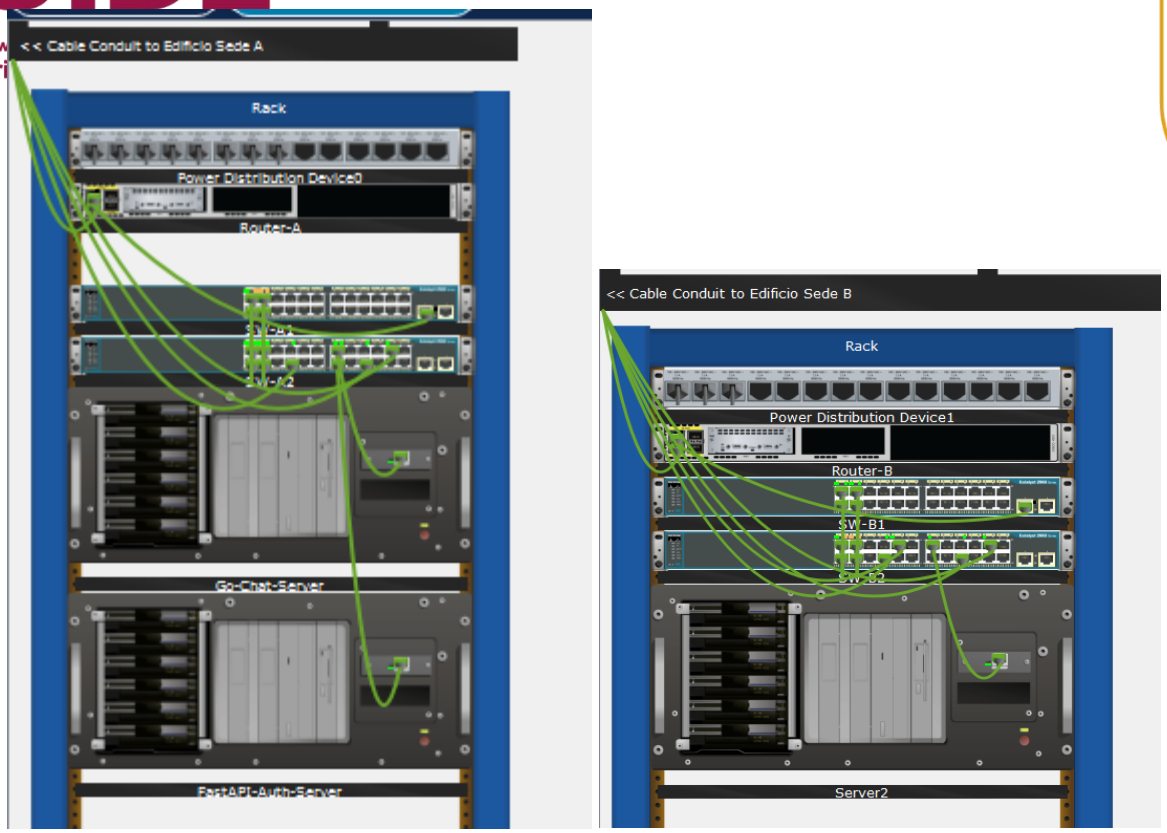
Internal distribution by floors and equipment rooms in Headquarters Building A and Headquarters Building B



Note. Architectural floor plan view of the buildings in Packet Tracer, detailing the location of the equipment room (wiring closet) and the physical layout of the cabling to end-user devices at each site.

Figure 3

Hardware implementation in the communications racks at Site A and Site B



Note. Detail of the physical assembly in the racks of the equipment rooms of both headquarters, showing the power distribution units, the perimeter routers (Router-A and Router-B), the pairs of redundant switches (SW-A1/SW-A2 and SW-B1/SW-B2) interconnected by EtherChannel, and the installed tower servers.

Logical Organization

The network's logical architecture has been designed to avoid a flat topology by implementing strict traffic control, segmentation-based security, and dynamic routing across the two sites. As the infrastructure is currently in the initial physical deployment phase, with no active device configurations, the planned logical model comprises the following technical pillars:

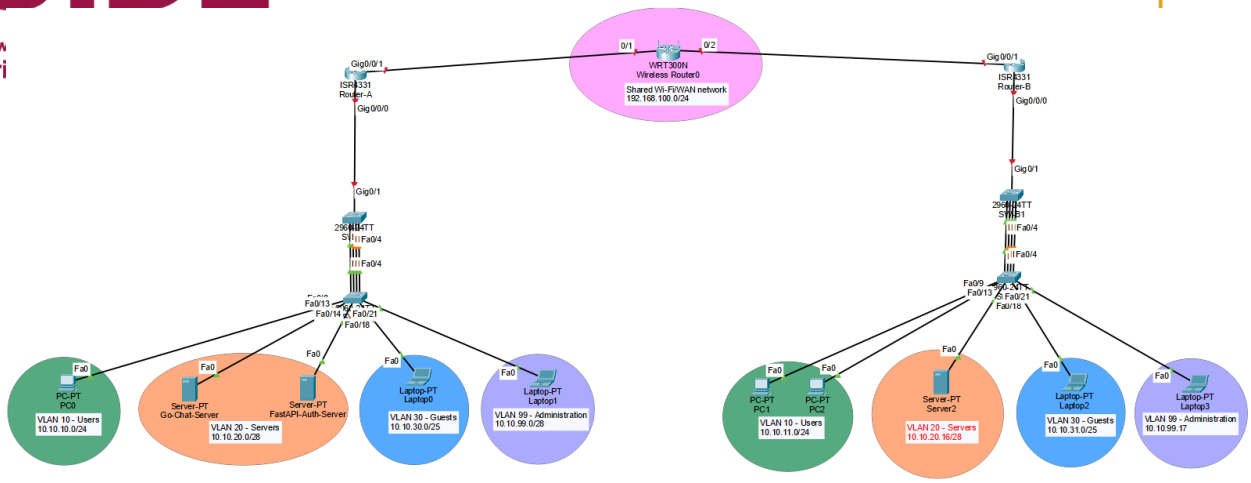
→ **VLAN segmentation:** Implementation of four identical, isolated VLANs at both sites to separate broadcast domains: VLAN 10 (Users and Collaborators), VLAN

20 (Dedicated servers for the Go messaging cluster and the FastAPI API), VLAN 30 (Guests), and VLAN 99 (Management and Administration).

- **Inter-VLAN routing (Router-on-a-Stick):** IEEE 802.1Q trunk links will be used from the core switches (SW-A1 and SW-B1) to the edge routers (Router-A and Router-B), allowing traffic between different VLANs to be routed in a controlled manner.
- **Redundancy and loop prevention (EtherChannel and RSTP):** To optimize traffic toward the server core and prevent Layer 2 broadcast storms, aggregated links using EtherChannel (LACP/PAgP)—consisting of four parallel physical ports—will be configured between the access switches at each site, complemented by the Rapid Spanning Tree Protocol (RSTP).
- **Dynamic routing (OSPFv2 in area 0):** The edge routers will communicate via the shared wireless network segment (192.168.100.0/24)—managed by the central Access Point (Wireless Router0)—running OSPFv2 in Area 0, enabling the subnets at Site B to exchange routes fully dynamically with the main data center at Site A.
- **Perimeter security (ACLs):** Access Control Lists (ACLs) will be applied to router interfaces under a strict implicit-deny policy, authorizing only traffic directed to the specific ports required by the Go and FastAPI services.

Figure 4

Diagrama of the planned logical topology with VLAN segmentation and a shared wireless WAN



Note. Logical view in Cisco Packet Tracer detailing the IP addressing assigned by subnet for each VLAN, the location of the Go and FastAPI servers at Site A, and the upstream interconnection core via Wireless Router0 (192.168.100.0/24), prior to executing configuration commands on the devices.

4. Addressing strategy

For the deployment of our distributed communications infrastructure, we have been assigned the private network block 10.10.0.0/16. To ensure compliance with scalability requirements (RNF-06), secure segmentation (RF-SEC-03), and comprehensive connectivity (RF-NET-01), we will apply a Variable Length Subnet Mask (VLSM) design.

The following sections detail the Subnet Allocation Worksheet and the Device Addressing Worksheet, designed to maintain a hierarchical structure and optimize IP address space.

Subnet Allocation Worksheet

Table 1

Subnet allocation worksheet for the distributed enterprise network (Site A and Site B)

Order	VLAN / Zone	Segment Name	Subnet / Prefix	First Usable IP	Last Usable IP	Broadcast
1	VLAN 10 (Site A)	Users (Site A)	10.10.10.0/24	10.10.10.1	10.10.10.254	10.10.10.255
2	VLAN 20 (Site A)	Go and FastAPI Servers (Site A)	10.10.20.0/28	10.10.20.1	10.10.20.14	10.10.20.15
3	VLAN 30 (Site A)	Guests (Site A)	10.10.30.0/25	10.10.30.1	10.10.30.126	10.10.30.127
4	VLAN 99	Administration (Site A)	10.10.99.0/28	10.10.99.1	10.10.99.14	10.10.99.15

	(Site A)					
5	VLAN 10 (Site B)	Remote Users (Site B)	10.10.11.0/24	10.10.11.1	10.10.11.254	10.10.11.255
6	VLAN 20 (Site B)	Branch Servers (Site B)	10.10.20.16/28	10.10.20.17	10.10.20.30	10.10.20.31
7	VLAN 30 (Site B)	Guests (Site B)	10.10.31.0/25	10.10.31.1	10.10.31.126	10.10.31.127
8	VLAN 99 (Site B)	Administration (Site B)	10.10.99.16/28	10.10.99.17	10.10.99.30	10.10.99.31
9	WAN	Shared Wireless WAN (OSPF Area 0)	192.168.100.0/24	192.168.100.1	192.168.100.254	192.168.100.255

Note. This table details the VLSM subnet allocation across the private block 10.10.0.0/16. It defines the network addresses, prefix lengths, usable IP ranges, and broadcast addresses for each departmental VLAN in Site A and Site B, as well as the point-to-point WAN link connecting Router A and Router B within OSPF Area 0.

Device Addressing Worksheet

Table 2

Device addressing and interface assignment worksheet

VLAN / Zone	Equipment	IP Assignment Type	Interface	IP Address	Subnet Mask
VLAN 10 (Site A)	PC0 (User device)	DHCP (Dynamic Pool)	Fa0	10.10.10.0	255.255.255.0
VLAN 10 (Site A)	Router A (Gateway)	Last Usable IP	g0/0/0.10	10.10.10.254	255.255.255.0
VLAN 20 (Site A)	Go-Chat-Server	2nd Usable IP	Fa0	10.10.20.2	255.255.255.240
VLAN 20 (Site A)	FastAPI-Auth-Server	3rd Usable IP	Fa0	10.10.20.3	255.255.255.240
VLAN 20 (Site A)	Router A (Gateway)	Last Usable IP	g0/0/0.20	10.10.20.14	255.255.255.240

VLAN 30 (Site A)	Laptop 0 (Guest device)	DHCP	Fa0	10.10.30.0	255.255.255.128
VLAN 30 (Site A)	Router A (Gateway)	Assigned Static IP	g0/0/0.30	10.10.30.65	255.255.255.128
VLAN 99 (Site A)	Laptop1 (Admin device)	DHCP	Fa0	10.10.99.0	255.255.255.240
VLAN 99 (Site A)	Router A (Gateway)	Assigned Static IP	g0/0/0.99	10.10.99.10	255.255.255.240
VLAN 10 (Site B)	PC1, PC2 (Remote users)	DHCP (Dynamic Pool)	Fa0	10.10.11.0	255.255.255.0
VLAN 10 (Site B)	Router B (Gateway)	Assigned Static IP	g0/0/0.10	10.10.11.100	255.255.255.0
VLAN 20 (Site B)	Server2 (Branch Node)	2nd Usable IP	Fa0	10.10.20.18	255.255.255.240
VLAN 20 (Site B)	Router B (Gateway)	Last Usable IP	g0/0/0.20	10.10.20.30	255.255.255.240
VLAN 30 (Site B)	Laptop2 (Guest device)	DHCP	Fa0	10.10.31.0	255.255.255.128
VLAN 30 (Site B)	Router B (Gateway)	Assigned Static IP	g0/0/0.30	10.10.31.50	255.255.255.128

VLAN 99 (Site B)	Laptop3 (Admin Device)	DHCP	Fa0	10.10.99.16	255.255.255.240
VLAN 99 (Site B)	Router B (Gateway)	Assigned Static IP	g0/0/0.99	10.10.99.25	255.255.255.240
WAN / OSPF	Router A (Gateway)	Assigned Static IP	g0/0/1	192.168.100.10	255.255.255.0
WAN / OSPF	Router B (Gateway)	Assigned Static IP	g0/0/1	192.168.100.11	255.255.255.0

Note. This worksheet specifies the exact IP assignment strategy, interface bindings, and subnet masks for every device and router subinterface across both sites. It details the static IP bindings reserved for backend servers (Go and FastAPI), the specific network base addresses mapped to DHCP dynamic pools for client workstations and laptops, and custom gateway IP addresses distributed across the usable ranges. Additionally, it maps the wireless interfaces connecting Router A and Router B to the shared wireless WAN infrastructure within OSPF Area 0 (192.168.100.0/24).

5. Integration plan

We have created an integration plan that specifies in detail where our software components will be located and how they will communicate over the proposed network topology to ensure safe and efficient operation of our business platform.

Operational location (Deployment)

Our distributed architecture is designed such that the full backend will be hosted centrally at Site A:

- The go messaging server and FastAPI API will be used as dedicated services in VLAN 20 (Isolated server network). This strategy means the logical core of the business is not exposed on the same network segment as the end users.
- Only the chat client (console/terminal) will be distributed and it will be running on the collaborators' workstations in Site A and Site B, in VLAN 10 (User Network).

Software level communication requirements

For the platform to function as an integrated ecosystem, the components will require the following communication flows:

- **Communication with FastAPI (Authentication):** When the user starts the chat client, the application sends a request (e.g., via HTTP/REST) to the IP address of the FastAPI server in VLAN 20. FastAPI will receive the credentials, check the security hashes in the database, and if they are correct, return an authorization status (token or confirmed session) to the client. This is communication that is short-lived and transactional.
- **Communication with Go (Concurrent messaging):** Once the client is authenticated, it will open a persistent connection through TCP Sockets directly to the server developed in Go, which is also in VLAN 20. The Go server will have to

keep these sockets open at the same time, listening, processing, and sending back text packets in real time between the different clients connected, wherever they are.

Communication infrastructure (network and security) support

To support these software flows, the network infrastructure needs to provide connectivity with strict control policies:

- **Inter-Site traffic:** The authentication requests and socket messages generated by the users of Site B will be routed through the local LAN switches (utilizing the EtherChannel bandwidth) to their perimeter router (Router-B). The OSPF dynamic routing protocol running over the shared wireless WAN segment (192.168.100.0/24) will then transmit the packets from the wireless link to the router at Site A (Router-A).
- **ACLs (Firewall policies):** The traffic will hit the perimeter router at Site A and try to enter VLAN 20 across the Router-on-a-Stick connection. This is where the internal firewall comes in. Access Control Lists (ACLs) will look at each packet and allow it through only if the traffic is destined to the specific TCP ports Go and FastAPI are listening to. Any other communication requests or scanning attempts to the servers will be blocked and implicitly discarded to guarantee the integrity of our application.

6. Work plan and responsibilities

Phase & Period	Specific Task / Activity	Associated Milestone	Dependencies	Responsible Member
Phase 1: Context, Planning and Organization (Weeks 1–2)	• Definition of business problem and organizational context. • Justified selection of the enterprise chat application. • Creation of Git repository, team contract, and licenses.	Milestone 1: Initial minutes and repository.	None (Start).	• Joselyn Laguna • Jose Velasco • Darwin Román
Phase 2: Analysis and Requirements (Weeks 3–5)	• Specification of network requirements (RF-NET) and sizing.	Milestone 2: Approved requirements specification.	Phase 1.	• Jose Velasco • Joselyn Laguna

	• Drafting of functional software requirements (RF-SW) and use cases. • Security requirements (RF-SEC) and privacy risk analysis. • Formulation of the quantitative proposal and preliminary instrument.			• Darwin Román • Joselyn Laguna
Phase 3: Solution Design (Weeks 6–7)	• Topology design in Cisco Packet Tracer, VLSM plan, and OSPFv2. • Preparation of Go app component diagrams and authentication sequence.	Milestone 3: Comprehensive architecture and design.	Phase 2.	• Joselyn Laguna • Jose Velasco • Darwin Román • Jose Velasco

	• FastAPI data model design, authorization policies, and firewall rules. • Test plan design and statistical metrics collection procedure.			
Phase 4: Prototype and Initial Development (Weeks 8–10)	• Initial configuration of Cisco switches (VLANs, Trunk, EtherChannel). • Modular development of the Go chat client and server. • Development of the FastAPI service and secure password hashing. • Execution of the pilot test for the statistical instrument.	Milestone 4: Initial functional prototype.	Phase 3.	• Darwin Román • Joselyn Laguna • Jose Velasco • Darwin Román

Phase 5: Integration and Deployment (Weeks 11–12)	• Integration and connection of the Go app with the FastAPI service on the network. • Final configuration of OSPF, NAT, DHCP, and router ACLs. • Inter-site connectivity tests and technical metrics logging.	Milestone 5: Integrated platform and initial deployment.	Phase 4.	• Jose Velasco • Joselyn Laguna • Darwin Román
Phase 6: Testing, Validation and Evaluation (Weeks 13–14)	• Execution of functional and security test case matrices. • Descriptive/inferential statistical analysis, tables, and graphs. • Elaboration of socio-technological, privacy, and ethics analysis.	Milestone 6: Validation and analysis report.	Phase 5.	• Jose Velasco • Darwin Román • Joselyn Laguna

Phase 7: Final Presentation and Defense (Week 15)	• Preparation and execution of the live technical demonstration. • Elaboration of the academic presentation and audiovisual support. • Final consolidation of the report and Git repository verification.	Milestone 7: Oral defense and demonstration.	Phase 6.	• Joselyn Laguna • Jose Velasco • Darwin Román
Phase 8: Closure and Final Delivery (Week 16)	• Correction and incorporation of tribunal observations. • Preparation of installation/execution manuals and final version tagging. • Consolidation of self-evaluations, peer evaluations, and closure.	Milestone 8: Final project delivery.	Phase 7.	• Joselyn Laguna • Jose Velasco • Darwin Román

7. Resources and Risks

Developing and implementing our distributed enterprise chat platform across the two sites requires a specific set of technological resources. Regarding hardware, we will use desktop and laptop computers for end-users, switches for local area networks, routers for the network perimeter, and devices running OpenWrt (such as Raspberry Pi or equivalent authorized platforms) for edge management and the shared wireless link.

As for software, the system will consist of a client-server application developed in Go to handle concurrent text messaging, supported by an authentication service built with FastAPI to ensure secure logins. Additionally, simulation tools like Cisco Packet Tracer will be used for network topology design, alongside database management systems for the persistent storage of credentials and logs.

The implementation environment will be set up in an emulated laboratory representing the two corporate sites (Site A and Site B). Both networks will be interconnected via OSPFv2 (Area 0) over a shared Wi-Fi/WAN network segment (192.168.100.0/24), utilizing the assigned 10.10.0.0/16 IP address block with subnetting via Variable Length Subnet Masking (VLSM), VLANs (VLAN 10 for users, VLAN 20 for servers, VLAN 30 for guests, and VLAN 99 for management), 802.1Q trunking, EtherChannel, RSTP, NAT, DHCP, and firewall rules.

During component development and integration, we may encounter potential technical difficulties, such as inter-site connectivity failures caused by OSPF dynamic routing errors, TCP socket interruptions during real-time message transmission, or issues validating tokens and

credentials via the FastAPI API. Additionally, there is a risk of unauthorized access or attempts to compromise the server network.

As proposed solutions, we will conduct thorough reviews of routing tables and perform connectivity tests using diagnostic commands (such as ping and traceroute) to mitigate network issues; for software failures, we will employ modular debugging and robust error handling in Go. Finally, to counter security risks, we will implement secure password hashing, Access Control Lists (ACLs) on routers, and strict firewall policies based on the principle of least privilege.

8. Validation plan

To ensure our enterprise chat platform operates optimally and meets required standards, we will implement a comprehensive validation plan that tests network infrastructure as well as software and security components. First, we will evaluate connectivity and routing using the `ping` and `traceroute` commands to verify IP communication between computers and subnets at both sites. We will also verify correct route propagation via the OSPFv2 protocol in Area 0 across the shared wireless link, as well as the effectiveness of Inter-VLAN routing (Router-on-a-Stick).

To test topology redundancy and stability, we will conduct tests using EtherChannel and Rapid Spanning Tree Protocol (RSTP). This will involve the controlled disconnection of one of the parallel physical links between switches to observe whether the network reconfigures its path transparently without interrupting data flow. Regarding service access, we will directly test the application by running the chat client, logging in from various workstations across both sites, and

sending concurrent messages to verify that the Go server processes information in real time and correctly stores message histories.

In terms of security policies, we will perform authentication tests by entering incorrect credentials and evaluating authorization mechanisms to ensure that the FastAPI service and firewall Access Control Lists (ACLs) block unauthorized attempts and protect sensitive resources. We will also verify that passwords are stored using secure hash encryption. Additionally, we will measure key performance indicators (KPIs) such as network latency, message delivery response time, and service availability.

Finally, as validation evidence, we will collect screenshots of command execution on router and switch CLIs, interface and service configuration records, traffic traces, logs from the Go and FastAPI applications, and the quantitative results obtained during testing. In this way, we will demonstrate in a reproducible manner that the project meets all established technical and security acceptance criteria.

9. Link to the project repository

https://github.com/Joselyn-uide/Chat-Empresarial-PI